PATRICK ORDISH ELEARNING

Make the inspection call.

A concise reference for this portfolio learning experience.

01 / Read the assembly

A 10.9 bolt has a tensile strength of 1000 N/mm^2. The .9 gives the ratio of yield strength to tensile strength. A Property Class 10 nut may be assembled with a 10.9 bolt. Check the complete specified assembly, manufacturer identification and documentation.

02 / Follow the combined method

Initial torque: set the wrench to 0.75 x the applicable reference torque, $M_{r,i}$. Complete this first stage for every bolt in the connection.
Reference mark: mark the nut position relative to the bolt threads.
Part turn: apply the specified rotation from that marked position.

Connected thickness	Rotation	Part turn
t < 2d	60 degrees	1/6 turn
2d <= t < 6d	90 degrees	1/4 turn
6d <= t <= 10d	120 degrees	1/3 turn

Keep the conditions. These values apply to Property Classes 8.8 and 10.9. t includes all packs and washers; d is the bolt diameter. Where the surface under the bolt head or nut, allowing for taper washers if used, is not perpendicular to the bolt axis, determine the required angle by testing. This reference does not cover thicknesses outside the stated ranges.

03 / Accept the evidence, not the appearance

Check completeness, alignment and the required tightening evidence. Resolve missing specified assemblies. Where an assembly is rejected, inspect the affected subgroup, correct the issue and reinspect. A photograph, a matching property class or a paint mark alone does not establish that every requirement has been met.

Learning reference only. Use the project specification and applicable standards for actual work. Adapted from the supplied structural-fastener training material: component markings; combined method, page 26, Table 21; connection and preloading inspection requirements. Retained source details are recorded separately in the course source map.